

Extending Transformers Represent Constrained Belief Geometry in the Residual Stream

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Abstract

Prior work shows that a single attention layer in a transformer trained on sequences from the Mess3 Hidden Markov process implements a *constrained belief update* in its residual stream, and that this geometry is linearly decodable from intermediate activations. We tried to study three questions left open: (i) how and when the constrained update gives way to the full Bayesian belief update as a function of the process parameters (α, x) ; (ii) what role the MLP layer plays in correcting the constrained approximation with higher-order terms; and (iii) how the belief state geometry and the transformer’s ability to recover it break down outside the parameter regime.

1 Background

1.1 Mealy Hidden Markov Chains

A *Mealy HMC* $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \pi, \{T^{(x)}\}_{x \in \mathcal{X}})$ consists of a vocabulary $\mathcal{X}$, a finite hidden-state set $\mathcal{S}$, a stationary distribution π , and sub-stochastic *emission-transition matrices* $T^{(x)}$ whose entries $T_{sd}^{(x)} = \Pr(x, s_d | s_s)$ give the joint probability of emitting symbol x and transitioning from state s_s to s_d . The matrices sum to a row-stochastic matrix: $\sum_x T^{(x)} = T$, with $\pi T = \pi$.

1.2 Mixed-State Presentation and Belief Updates

An observer who cannot see the hidden state maintains a *belief* $\eta \in \Delta^{|\mathcal{S}|-1}$, the posterior distribution over states. Starting from $\eta_0 = \pi$, observing token x updates the belief via

$$\eta \leftarrow \frac{\eta T^{(x)}}{\eta T^{(x)} \mathbf{1}}, \quad (1)$$

where $\mathbf{1}$ is the all-ones column vector. The denominator $\eta T^{(x)} \mathbf{1} = \Pr(x | \eta)$ is the *state-dependent* emission probability. The set of all reachable beliefs $\mathcal{R} \subseteq \Delta^{|\mathcal{S}|-1}$, together with the dynamics (1), constitutes the *Mixed-State Presentation* (MSP) - the minimal sufficient statistic for optimal prediction [2]. For the processes we study, $\mathcal{R}$ is generically an infinite fractal set.

Constrained belief update. Piotrowski et al. [1] show empirically and analytically that a single-layer transformer attention head implements a linearized approximation:

$$r_d^{\text{con}} = \pi + \sum_{i=1}^d (\pi S^{(z_i)} T^{d-i} - \pi), \quad T = \sum_x T^{(x)}, \quad (2)$$

where $S_{sd}^{(z)} = T_{sd}^{(z)} / p_s^z$ is the row-normalized transition matrix ($p_s^z = \sum_d T_{sd}^{(z)}$ is the marginal emission probability of symbol z from state s).

*Code: <https://github.com/ilatims-b/extending-constrained-belief>

2 The Mess3 Process

2.1 Definition

Mess3 is a Mealy HMC with three hidden states $\{A, B, C\}$, three symbols $\{0, 1, 2\}$, and two scalar parameters $\alpha \in (0, 1)$, $x \in (0, \frac{1}{2})$. Define $b = (1 - \alpha)/2$ and $y = 1 - 2x$. The emission-transition matrices are

$$T^{(0)} = \begin{pmatrix} \alpha y & bx & bx \\ \alpha x & by & bx \\ \alpha x & bx & by \end{pmatrix}, \quad (3)$$

with $T^{(1)}$, $T^{(2)}$ obtained by cyclically permuting states. Symbol z has a *dominant* state (the state most likely to emit it): A for $z = 0$, B for $z = 1$, C for $z = 2$.

Every entry of $T^{(0)}$ (and its cyclic permutations) is a product of α, b, x, y , each strictly positive for $\alpha \in (0, 1)$, $x \in (0, \frac{1}{2})$.

2.2 Emission Probabilities

By the $\mathbb{Z}_3$ symmetry of Mess3, only two distinct marginal emission probabilities exist:

$$p_{\text{dom}} = \alpha y + 2bx \quad (\text{dominant state}), \quad (4)$$

$$p_{\text{off}} = \alpha x + b(1 - x) \quad (\text{off-dominant states}). \quad (5)$$

One verifies $p_{\text{dom}} + 2p_{\text{off}} = 1$ (row sums of Eq. (3) and its cyclic images). Their difference,

$$\Delta p = p_{\text{dom}} - p_{\text{off}} = \frac{(3\alpha - 1)(1 - 3x)}{2}, \quad (6)$$

is the single scalar that turns out to organize every phenomenon studied in this paper: it vanishes on two lines, $\alpha = \frac{1}{3}$ and $x = \frac{1}{3}$, and is strictly positive on two disjoint open regions, (I) $\alpha > \frac{1}{3}$, $x < \frac{1}{3}$ and (II) $\alpha < \frac{1}{3}$, $x > \frac{1}{3}$.

2.3 Row-Normalized Matrices

The matrix $S^{(z)}$ with entries $S_{sd}^{(z)} = T_{sd}^{(z)}/p_s^z$ is row-stochastic and represents the *conditional* next-state distribution given emission z . For LinearMess3 (used in Section 6), the data-generating step factorizes as: first sample $z \sim p_s^z$, then sample the next state from $S^{(z)}$. Since $p_s^z \cdot S_{sd}^{(z)} = T_{sd}^{(z)}$, the joint distribution over (z, s_{next}) is identical to standard Mess3, so *LinearMess3* and *Mess3* generate statistically identical sequences (immediate from the definition of $S^{(z)}$) but the belief update is different.

2.4 Closed Form for T

We record here a fact used repeatedly below (Appendix C gives the derivation): by direct computation from Eq. (3) and its cyclic permutations, $T = \sum_z T^{(z)}$ has the closed form

$$T = (1 - 3x)I + xJ, \quad (7)$$

where I is the 3×3 identity and J the all-ones matrix. Consequently, for any row vector v with $v\mathbf{1} = 0$ (a “zero-sum” vector), $vJ = (v\mathbf{1})\mathbf{1}^\top = 0$, so

$$vT = (1 - 3x)v \quad \text{exactly, for every zero-sum } v. \quad (8)$$

T acts as the *exact scalar* $1 - 3x \in (0, 1)$ on the zero-sum subspace, this is stronger than a mere contraction bound, and is used in Appendix D and Section 6 below.

3 Three Belief Updates

We now have three candidate descriptions of what the transformer’s residual stream might encode after d tokens: the exact Bayesian posterior, the linear approximation to it, and the attention-only constrained update of prior work.

Let $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and sequence $z_1, \dots, z_d$.

Full Bayesian update.

$$\eta^{(d)} = \frac{\pi T^{(z_1)} \dots T^{(z_d)}}{\pi T^{(z_1)} \dots T^{(z_d)} \mathbf{1}}. \quad (9)$$

Linear update.

$$r_d^{\text{lin}} = \pi S^{(z_1)} S^{(z_2)} \dots S^{(z_d)}. \quad (10)$$

No renormalization needed: r_d^{lin} is automatically a probability vector. This is the update we introduce and study here: unlike the full Bayesian update, each step only ever multiplies by a fixed row-stochastic matrix, so it is linear in the belief at every step, (i.e. over time/ tokens).

Constrained update (attention). As in Eq. (2).

Remark 3.1. The constrained and linear updates coincide at $d = 1$ from $\pi : r_1^{\text{con}} = \pi + (\pi S^{(z_1)} T^0 - \pi) = \pi S^{(z_1)} = r_1^{\text{lin}}$, using $T^0 = I$.

4 When Does the Linear Update Equal the Full Update?

4.1 The Key Observation

The full update Eq. (9) and the linear update Eq. (10) differ only because the denominator of (9) depends on η . After one step from π :

$$\eta_d^{(1)} = \frac{1}{3} \sum_s T_{sd}^{(z)}, \quad (11)$$

$$r_{1,d}^{\text{lin}} = \frac{1}{3} \sum_s \frac{T_{sd}^{(z)}}{p_s^z}. \quad (12)$$

These are equal if and only if p_s^z is the same for all states s , so that it factors out of the sum in (12).

4.2 Exact Condition

Setting $p_{\text{dom}} = p_{\text{off}}$ and using Eqs. (4)–(5):

$$(\alpha - b)(1 - 3x) = 0, \quad (13)$$

which holds in exactly two cases.

Theorem 4.1 (Exact equality). *The linear update equals the full Bayesian update for all η and all sequence lengths d if and only if $\Delta p = 0$, i.e.*

$$\alpha = \frac{1}{3} \quad \text{or} \quad x = \frac{1}{3}.$$

On both lines $p_s^z = \frac{1}{3}$ for all states and symbols: every token is uninformative about the hidden state.

Proof. ($\Rightarrow$, necessity) If the two updates agree for all η and d , they agree in particular at $d = 1$, $\eta = \pi$: by Eqs. (11)–(12), this forces p_s^z to be independent of s for every z , i.e. $p_s^z = \frac{1}{3}$ for all s, z (since the two values realized by p_s^z average to $\frac{1}{3}$ by $p_{\text{dom}} + 2p_{\text{off}} = 1$). By Eq. (6) this is exactly $\Delta p = 0$.

($\Leftarrow$, sufficiency) Suppose $\Delta p = 0$. Then $p_{\text{dom}} = p_{\text{off}} = \frac{1}{3}$, so $p_s^z = \frac{1}{3}$ for *every* state s and symbol z - not only at $\eta = \pi$, but as a structural property of the matrices $T^{(z)}$ themselves. Consequently, for *any* $\eta \in \Delta^2$ and any symbol z ,

$$\eta T^{(z)} \mathbf{1} = \sum_s \eta_s p_s^z = \frac{1}{3} \sum_s \eta_s = \frac{1}{3},$$

so the full-update denominator is the constant $\frac{1}{3}$ regardless of η , and

$$\eta \mapsto \frac{\eta T^{(z)}}{\eta T^{(z)} \mathbf{1}} = 3 \eta T^{(z)} = \eta S^{(z)}$$

(using $S^{(z)} = T^{(z)}/p_s^z = 3T^{(z)}$ row-wise when $p_s^z \equiv \frac{1}{3}$). Thus the full-update map and the linear-update map $\eta \mapsto \eta S^{(z)}$ coincide as *functions on the entire simplex*, not merely at a single point. Since both updates start from the same $\eta^{(0)} = r_0^{\text{lin}} = \pi$ and apply the same sequence of identical maps, induction on d gives $\eta^{(d)} = r_d^{\text{lin}}$ for every d . $\square$

Away from these two lines, the two updates differ, by an amount that depends on (α, x) and grows with d . Next we characterize this gap and ask which of the two updates the transformer actually tracks, which is the central empirical claim of this paper.

5 Does the Transformer Track the Linear or the Full Bayesian Update?

Theorem 4.1 pins down the two lines where linear and full Bayesian coincide exactly. Everywhere else, they diverge, and we would like a theoretical handle on *how much*, so that a comparison against the transformer’s actual residual stream can be read at parameter settings spanning a whole range of divergence - from nearly-identical to substantially apart. Hence, we come up with an error bound between the two updates, then uses it to organize the empirical comparison.

Define the *contrast ratio* $\rho = \max(p_{\text{dom}}, p_{\text{off}}) / \min(p_{\text{dom}}, p_{\text{off}}) \geq 1$ (equal to $p_{\text{dom}}/p_{\text{off}}$ on regime (I), the branch used throughout). Note $\rho = 1 \Leftrightarrow \Delta p = 0$ (Theorem 4.1).

5.1 One-step bound

Lemma 5.1 (One-step error bound). *Starting from $\eta^{(0)} = r_0^{\text{lin}} = \pi$,*

$$\|\eta^{(1)} - r_1^{\text{lin}}\|_1 \leq \frac{2(\rho - 1)}{\rho}. \quad (14)$$

Proof. In Mess3, p_s^z takes only two values: p_{dom} if s is the dominant state for z , and p_{off} otherwise. From Eqs. (11)–(12),

$$r_{1,d}^{\text{lin}} = \frac{1}{3} \sum_s T_{sd}^{(z)} / p_s^z \geq \frac{1}{3} \sum_s T_{sd}^{(z)} / p_{\text{dom}} = \eta_d^{(1)} / p_{\text{dom}}, \quad r_{1,d}^{\text{lin}} \leq \eta_d^{(1)} / p_{\text{off}}.$$

Hence, for every d ,

$$\frac{\eta_d^{(1)}}{\rho} \leq r_{1,d}^{\text{lin}} \leq \rho \eta_d^{(1)} :$$

the two probability vectors are *mutually dominated* with ratio ρ . For any two probability vectors p, q with $\frac{1}{\rho}p \leq q \leq \rho p$ entrywise, and every index k , $\min(p_k, q_k) \geq \max(p_k, q_k)/\rho$, so

$$1 - TV(p, q) = \sum_k \min(p_k, q_k) \geq \frac{1}{\rho} \sum_k \max(p_k, q_k) \geq \frac{1}{\rho},$$

i.e.

$$TV(p, q) \leq \frac{\rho - 1}{\rho}, \quad \text{and} \quad \|p - q\|_1 = 2TV(p, q) \leq \frac{2(\rho - 1)}{\rho}.$$

Applying this with $p = \eta^{(1)}$, $q = r_1^{\text{lin}}$ gives the claim. $\square$

5.2 Multi-step generalization

Lemma 5.1 suggests the natural conjecture

$$\|\eta^{(d)} - r_d^{\text{lin}}\|_1 \leq \frac{2(\rho - 1)}{\rho} \cdot \frac{1 - (1 - 3x)^d}{3x} \quad \text{for every } d. \quad (15)$$

For general d , a telescoping argument carries the proof through every step except one – bounding a “starting-point” contribution to the per-step error, which turns out not to admit a bound of the needed form. Proof attempt is written in Appendix D.

5.3 Numerical evidence

Exhaustive search over all 3^d token sequences for $d \leq 7$, at parameter points spanning the regime studied here (including the five points of Table 1), never violated Eq. (15): the true error stayed within roughly 1–40% of the bound in every case. Table 1 and Figure 1 give the bound’s value across the parameter space.

Table 1: Regimes for the error bound of Eq. (15).

Regime	Condition	Bound behavior
Exact equality	$\alpha = \frac{1}{3}$ or $x = \frac{1}{3}$	$= 0$
Near equality	$\Delta p \approx 0$	≈ 0
Short sequences	$d \ll 1/(3x)$	$\approx d \cdot (\rho - 1)/\rho$
Fast mixing	$x \rightarrow \frac{1}{3}$	both factors $\rightarrow 0$
Worst case	$x \rightarrow 0, \alpha \rightarrow 1$	$\rightarrow \infty$

5.4 The main empirical result

Empirically, transformers represent the linear belief update better than the full Bayesian belief update in the residual stream.

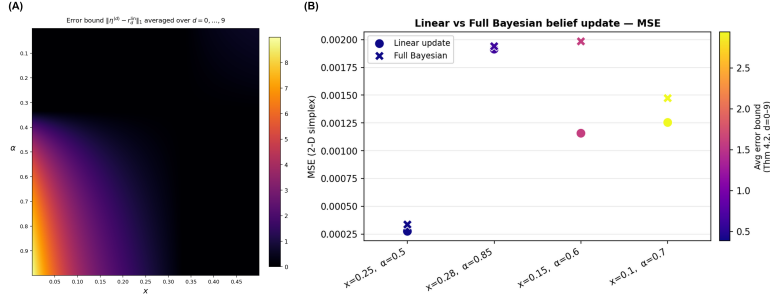


Figure 1: Error bound between the Linear and Full Bayesian updates, and the residual stream’s empirical preference between them. (A) Error bound (15) averaged over $d = 0, \dots, 9$ as a function of the Mess3 parameters (α, x) . Bright regions (top-left, small x , large α) correspond to large ρ and long mixing times; the two exact-equality lines $\alpha = \frac{1}{3}$ and $x = \frac{1}{3}$ appear as dark bands where the bound vanishes. (B) Empirical comparison of Mean Squared Error (MSE) between the transformer residual stream and the theoretical Linear vs. Full Bayesian updates for different parameterisations. Marker color represents the theoretical error bound from (A) for that Mess3 (x, α) ; models were chosen across different regimes of error bound for a fair comparison. The result holds true across all these models.

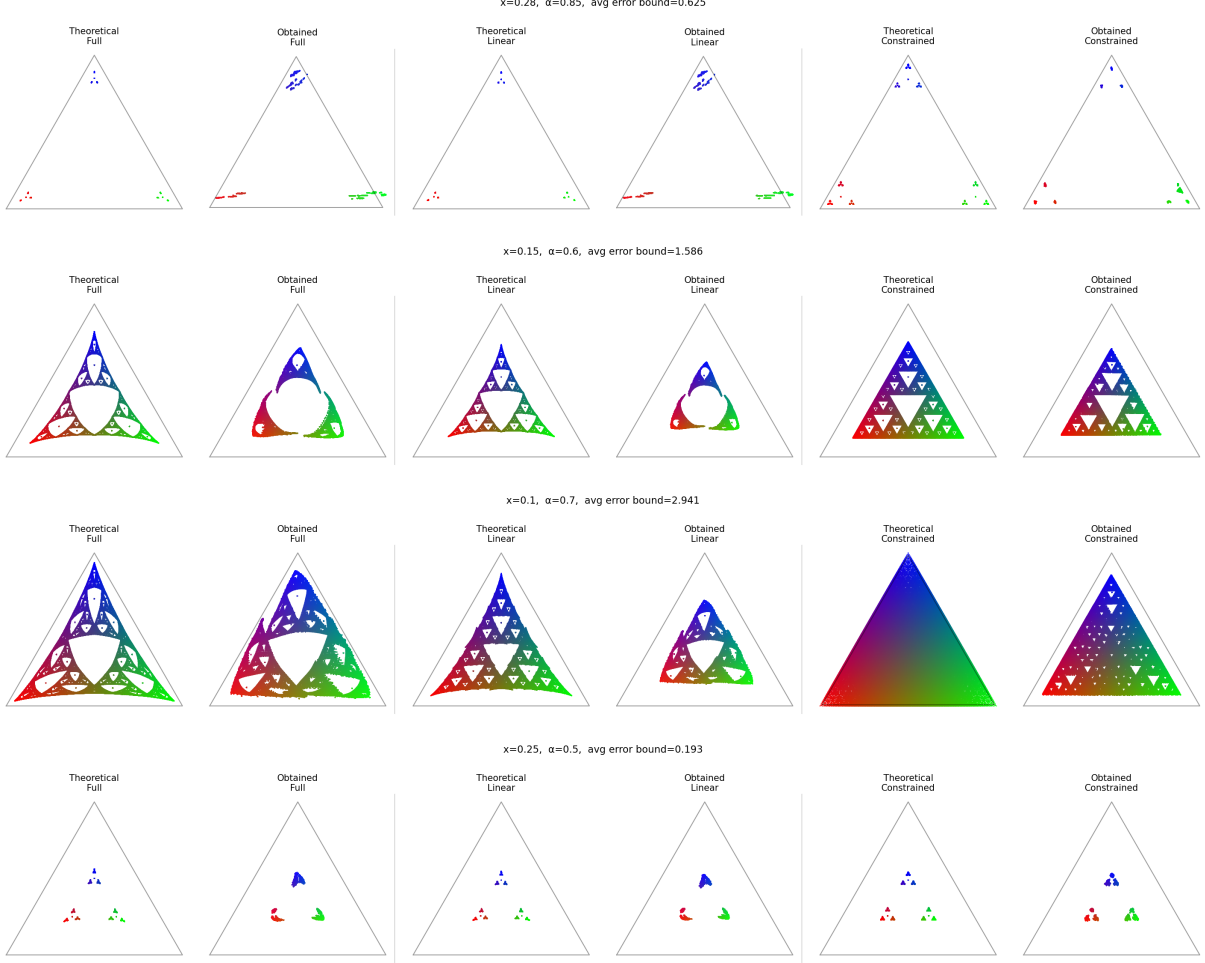


Figure 2: Simplex galleries comparing Theoretical and Obtained beliefs across four parameterisations.

In the regime of parametrization considered for the constrained update, the full Bayesian and linear belief updates are close (small Δp). Since the transformer represents Linear belief update and the MLP is hard to probe directly against the full Bayesian update because of its non-linearity, we work with the linear update throughout the rest of this paper.

6 What Does the MLP Do?

Since r_d^{con} is only the attention layer’s contribution, this raises a natural question: what is the MLP doing? Before answering it, we address a prior puzzle. r_d^{con} is by construction *linear* in the per-step deviations Δ_i (defined below) – but attention is built from a *nonlinear* softmax. So does the softmax collapses to something effectively linear? And turns out it does (Section 6.1). Sections 6.2 and 6.3 then turn to the MLP itself.

6.1 Why Softmax Behaves Linearly

The constrained update requires attention weights of the form

$$A_{d,s} = m_d \lambda^{d-s}, \quad \lambda = 1 - 3x \in (0, 1), \quad (16)$$

for constants m_d . Since $A = \text{softmax}(L)$ for logits L ,

$$\ln A_{d,s} = L_{d,s} - C(d) \quad (17)$$

for a normalizing constant $C(d)$ depending only on d (not s); substituting the assumed form for $A_{d,s}$ gives

$$L_{d,s} = -s \ln \lambda + f(d), \quad (18)$$

where $f(d) = \ln m_d + d \ln \lambda + C(d)$ absorbs everything d -dependent, i.e. the pre-softmax logit is linear in the *source* position s (with an arbitrary d -dependent offset, eliminated by softmax’s translation invariance). So the required attention pattern needs only a linear logit, which a single linear query/key computation can produce; softmax’s nonlinearity does not stand in the way, because it only ever acts on a one-dimensional linear family of logits, not on the belief itself.

At $(x, \alpha) = (0.15, 0.6)$, fitting $L_{d,s} = -s \ln \lambda + f(d)$ to the trained model’s pre-softmax logits, row by row (fixed destination d , varying source s), gives a consistently good linear fit and a slope close to the predicted $-\ln \lambda = -\ln(1 - 3x) \approx 0.598$ (Table 2): the mean fitted slope across $d = 2, \dots, 9$ is 0.6124 ($R^2 \geq 0.98$ for $d \leq 7$, degrading mildly at $d = 8, 9$ as context grows scarce), corresponding to an empirical $\lambda \approx 0.542$ against the theoretical 0.55.

Table 2: Row-wise linearity check of the pre-softmax logit $L_{d,s} = -s \ln \lambda + f(d)$ at $(x, \alpha) = (0.15, 0.6)$, fixing destination d and regressing over source s . Theory predicts slope $-\ln \lambda = -\ln(1 - 3x) \approx 0.598$.

d	Slope $(-\ln \lambda)$	R^2	Intercept $f(d)$
2	0.6242	1.0000	-1.5412
3	0.6200	0.9987	-1.4119
4	0.6293	0.9996	-1.7243
5	0.6823	0.9904	-1.1597
6	0.6568	0.9987	-2.2919
7	0.6030	0.9807	-2.1711
8	0.5989	0.9763	-2.5456
9	0.4843	0.8895	-2.4437
Mean slope: 0.6124		Empirical λ : 0.5421	

6.2 The Higher-Order Residual

Given that attention can implement r_d^{con} , what closes the remaining gap to r_d^{lin} ? Define $\Delta_i = S^{(z_i)} - T$ for each position i ; note $\Delta_i \mathbf{1} = 0$ (both $S^{(z_i)}$ and T are row-stochastic). The constrained update (2) collects zeroth- and first-order terms in the Δ_i :

$$r_d^{\text{con}} = \pi + \sum_{i=1}^d \pi \Delta_i T^{d-i}. \quad (19)$$

The linear update (10) expands as a product:

$$r_d^{\text{lin}} = \pi \prod_{i=1}^d (T + \Delta_i). \quad (20)$$

Expanding the product, the zeroth- and first-order terms in (20) reproduce (19) exactly. The remainder consists of cross-products of two or more Δ_i . For $d = 1, 2, 3$:

$$d = 1. \quad r_1^{\text{lin}} = \pi(T + \Delta_1) = \pi T + \pi \Delta_1 = \pi + \pi \Delta_1 = r_1^{\text{con}} \text{ (using } \pi T = \pi \text{)}$$

$$d = 2.$$

$$\begin{aligned} r_2^{\text{lin}} &= \pi(T + \Delta_1)(T + \Delta_2) = \pi T T + \pi T \Delta_2 + \pi \Delta_1 T + \pi \Delta_1 \Delta_2 \\ &= \pi + \pi \Delta_2 + \pi \Delta_1 T + \pi \Delta_1 \Delta_2 \quad (\pi T T = \pi T = \pi, \quad \pi T \Delta_2 = (\pi T) \Delta_2 = \pi \Delta_2) \\ &= r_2^{\text{con}} + \pi \Delta_1 \Delta_2. \end{aligned}$$

One cross-term: $S = \{1, 2\}$.

$d = 3$.

$$\begin{aligned} r_3^{\text{lin}} &= \pi(T + \Delta_1)(T + \Delta_2)(T + \Delta_3) \\ &= r_3^{\text{con}} + \pi\Delta_1\Delta_2T + \pi\Delta_1\Delta_3T + \pi\Delta_2\Delta_3 + \pi\Delta_1\Delta_2\Delta_3, \end{aligned}$$

keeping every T in its literal expansion position: the $S = \{1, 2\}$ and $S = \{1, 3\}$ terms both carry a trailing T (since $\max S < 3$ for neither - $\max\{1, 2\} = 2 < 3$, and the $S = \{1, 3\}$ term's T literally sits between Δ_1 and Δ_3 in the expansion, $\pi\Delta_1T\Delta_3$, which we still write as $\pi\Delta_1\Delta_3T$ below since the two are equal, though not for a reason we use yet), while $S = \{2, 3\}$ (with $\max S = 3 = d$) and $S = \{1, 2, 3\}$ carry none. Four cross-terms in total. (Verified numerically at $(\alpha, x) = (0.6, 0.15)$ for $z_1, z_2, z_3 = 0, 1, 2$: all four terms and their sum match the directly computed $r_3^{\text{lin}} - r_3^{\text{con}}$ to floating-point precision.)

General pattern. One term per subset $S \subseteq \{1, \dots, d\}$ with $|S| \geq 2$: Δ_i at each position $i \in S$, T elsewhere, in original left-to-right order. The number of such subsets is $2^d - d - 1$ (total subsets 2^d , minus the $d + 1$ subsets of size ≤ 1) - exponential in d .

A closed form (optional simplification). The T 's in each term can be collapsed to a single scalar. Matrix multiplication does not commute in general, so the position of each T relative to the Δ_i 's matters - but because $\Delta_i J = 0$ (zero row sums) and $T = (1 - 3x)I + xJ$ (Eq. (7)), $MT = (1 - 3x)M$ exactly for *any* matrix M with zero row sums, in particular any partial product of Δ_i 's. Consequently every T appearing after position $\min(S)$ (sandwiched between two Δ_i 's or trailing after $\max(S)$, as in the $d = 3$ terms above) collapses to a scalar factor $(1 - 3x)$:

$$r_d^{\text{lin}} = r_d^{\text{con}} + \underbrace{\sum_{|S| \geq 2} (1 - 3x)^{d - \min S + 1 - |S|}}_{\text{higher-order residual}} \pi \prod_{i \in S} \Delta_i, \quad (21)$$

where the exponent counts the T 's among positions $\min S, \dots, d$ not in S - including internal gaps, not only trailing T 's: for $d = 3$, $S = \{1, 3\}$ has one internal gap (position 2) and no trailing T ($\max S = 3 = d$), giving exponent 1, matching $\pi\Delta_1T\Delta_3 = \pi\Delta_1\Delta_3T = (1 - 3x)\pi\Delta_1\Delta_3$ above. A formula using only $T^{d - \max S}$, with no allowance for internal gaps, would instead give exponent 0 for this term - *not* correct for non-consecutive S .

6.3 Hypothesis: The MLP Supplies the Correction

We hypothesize - but have **not yet verified empirically** - that the MLP supplies the higher-order cross-terms of Eq. (21), bridging r_d^{con} to r_d^{lin} :

$$\text{MLP}(\text{LN}(r_d^{\text{con}})) \stackrel{?}{\approx} r_d^{\text{lin}} - r_d^{\text{con}}. \quad (22)$$

This is directly testable: regress the MLP's actual output against the target $r_d^{\text{lin}} - r_d^{\text{con}}$ and check the residual. We report the hypothesis and the derivation that makes it precise, and leave the empirical check to future work (Section 9).

The number of cross-terms in Eq. (21) grows as $2^d - d - 1$ (the count of subsets with $|S| \geq 2$, unaffected by the exponent correction above) - exponentially in the sequence length d . If the MLP really is computing something like this correction, representing more of these terms exactly should require more residual-stream capacity as d grows, i.e. a larger d_{model} ; we would also expect some kind of saturation or convergence in how well a fixed-width MLP can approximate the sum as d increases; since each individual term shrinks (data dependent, since it is a product of d increasingly many Δ_i 's, each a bounded matrix). We do not explore either question further here.

7 Training Dynamics at an Extreme Point: SGD vs. Adam

Separately from the linear-vs-full-Bayesian question, we trained transformers at an extreme parameter point, $(x, \alpha) = (0.01, 0.01)$, far from the regime used above. At this point $\Delta p \approx -0.47$: it lies in neither regime (I) nor regime (II) of Section 2 (both require $\Delta p > 0$), so the results of Section 8 and Appendix B are not directly established here - we expect the same qualitative corner-concentration mechanism to apply, by a symmetry argument swapping which state counts as "dominant."

Figure 3 compares the theoretical belief simplex against what SGD- and Adam-trained models actually recover; the two optimizers converge to visibly different geometry from identical training data. SGD’s recovered simplex covers essentially all theoretically reachable mixed states, matching the full (unique-states) theoretical simplex; Adam’s recovered simplex instead concentrates on the high-probability corner region, much closer to the frequency-weighted (Blackwell) theoretical simplex than to the full one. Training curves (loss, simplex MSE, KL divergence) are given in Appendix F.

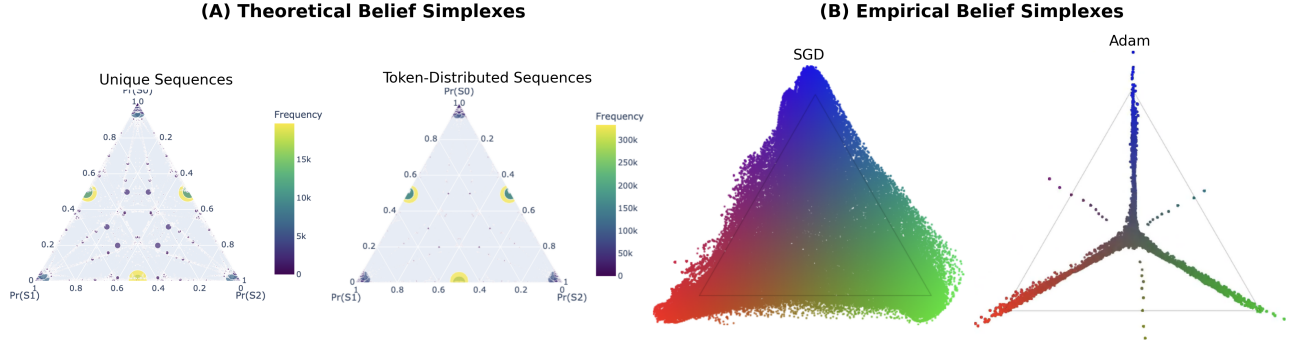


Figure 3: Theoretical vs. retrieved simplexes for SGD and Adam at $(x, \alpha) = (0.01, 0.01)$. (A) Theoretical belief simplexes when considering all unique reachable states versus sequences sampled from the actual distribution. (B) Empirical simplexes retrieved by SGD and Adam. SGD represents all reachable mixed states; Adam covers mostly high-mass (corner) beliefs.

Why would two optimizers trained on the same data end up representing different geometry? Section 8 develops the missing-mass framework that tries answer this: the training *distribution* itself (frequency-weighted sequences) is not the same as the set of geometrically distinct states, and how much of the gap between them an optimizer closes appears to depend on the optimizer.

8 Missing Mass and the Blackwell Measure

8.1 Two Measures on the Same Simplex

When a transformer is trained on N sampled sequences, the distribution of mixed states it sees is the *Blackwell measure* μ_B i.e. the stationary measure of the MSP weighted by sequence frequency (Appendix A gives the precise definition). A complementary distribution μ_{uniq} instead assigns equal weight to each *geometrically distinct* mixed state, regardless of how often it is visited. These are the two theoretical simplexes compared in Figure 3: SGD’s recovered geometry looks more like μ_{uniq} ’s support, Adam’s more like μ_B ’s.

The *missing mass* -probability assigned to unvisited sequences is estimated in Appendix A by an asymptotic scaling argument (heuristically) to decay as

$$M_N \sim \mathcal{O}\left(\left(\frac{p_{\text{off}}}{p_{\text{dom}}}\right)^n\right) = \mathcal{O}\left(\left(\frac{1 - \Delta p}{1 + 2\Delta p}\right)^n\right), \quad (23)$$

large (slow decay) when Δp is large: on regime (I), a large gap between p_{dom} and p_{off} means most probability mass concentrates on short, low-diversity words, so most of the geometrically distinct interior states are rarely or never seen during training, however large N is. A related heuristic estimate gives $\sigma_{\text{uniq}}^2 / \sigma_{\text{seq}}^2 \sim e^{2n|\lambda_{\text{Lyap}}|}$, where $\lambda_{\text{Lyap}} < 0$ is the Lyapunov exponent of the IFS [3] (Appendix A). Both make precise the same qualitative picture as Figure 3: μ_B concentrates near the corners while the full set of geometrically distinct reachable states spans the interior. If an optimizer’s implicit bias favors fitting the frequency-weighted training signal, it should recover something closer to μ_B (Adam, in Figure 3); if it instead manages to represent rarely-visited states too, it recovers something closer to μ_{uniq} (SGD). We are not sure what it means for Adam and SGD to represent different simplexes in residual stream,

8.2 Direct Evidence Inside Regime (I)

Section 7’s illustration sits outside regime (I) ($\Delta p < 0$ at $(0.01, 0.01)$), so Appendix B’s monotonicity mechanism (reason why corner concentration happens) does not strictly, apply to it. Inside regime (I), where it does apply, we have direct trained-model evidence at four points, spanning $\Delta p = 0.06$ to 0.39 : Figure 2 (Section 5.4) compares, at each point, the full theoretical simplex against the simplex a trained model actually recovers. This is the same theoretical-vs-obtained comparison as Figure 3, now at points where Lemma B.1’s monotonicity - $f^{(z)}$ pushes $\eta_{s^*}(z)$ upward on every step, with the exact rate given there.

8.3 Two Phenomena, One Scalar

Table 3 summarizes the two phenomena this section and Section 5 develop, both as functions of the same scalar Δp : corner concentration of the Blackwell measure (captured by two related quantities, the missing mass and the variance ratio) and the

Table 3: Two phenomena controlled by $\Delta p = (3\alpha - 1)(1 - 3x)/2$.

Phenomenon	Large Δp	Small Δp
Missing mass M_N	High	Low
σ_{seq} (Blackwell)	Low (corners)	High (spread)
Linear vs. full update error	Large	Zero (Theorem 4.1)

9 Discussion and Future Work

Closing the multi-step error-bound gap. Completing a valid proof of the multi-step generalization (Section 5.2, Appendix D) is the most direct open work from Section 5. Numerical evidence (exhaustive search, $d \leq 7$) supports it strongly; what is missing is a correct bound on the “starting-point error” $(\eta^{(i-1)} - r_{i-1}^{\text{lin}})\Delta_i$ that does not fall back on the operator-norm inequality that failed.

Testing the MLP hypothesis. The hypothesis that the MLP computes (an approximation to) the higher-order residual $r_d^{\text{lin}} - r_d^{\text{con}}$ - has not been checked against actual model internals. The direct test is a regression of MLP output against this target, across sequence lengths d and the same range of (α, x) used in Section 5. A related open question is whether representing more of the $2^d - d - 1$ cross-terms as d grows requires d_{model} to grow with it, and whether there is a capacity at which the MLP’s approximation saturates or converges.

The SGD/Adam gap, and the branch outside regime (I). is reported as an empirical observation; we have not derived why the two optimizers differ in which part of the simplex they represent, only connected the phenomenon to the missing-mass framework qualitatively. That parameter point also lies outside regime (I) entirely ($\Delta p < 0$), so a proper extension of Appendix B to $\Delta p < 0$ (and to regime (II), where the corner-attraction mechanism reverses) would be needed.

Extended regimes. Beyond the above, extending the analysis to processes with more hidden states, or to nonunifilar HMCs with richer overlap structure, is a natural further step not attempted here.

References

References

- [1] M. Piotrowski et al., *Transformers Represent Constrained Belief State Geometry in their Residual Stream*, arXiv:2502.01954 (2025).
- [2] J. P. Crutchfield, *The Calculi of Emergence: Computation, Dynamics, and Induction*, Physica D **75**, 11–54 (1994).

- [3] A. Jurgens and J. P. Crutchfield, *Divergent Predictive States: The Statistical Complexity Dimension of Stationary, Ergodic Hidden Markov Processes*, Chaos **31**, 083114 (2021).
- [4] A. Jurgens and J. P. Crutchfield, *Shannon Entropy Rate of Hidden Markov Processes*, J. Stat. Phys. **183**, 32 (2020).

A Missing Mass, Blackwell Measure, and Lyapunov Exponent

A.1 Two Distributions Over Mixed States

Let $\mathcal{W}_n$ denote all length- n words over $\{0, 1, 2\}$. Every word $w = z_1 z_2 \cdots z_n$ has a probability $P(w)$ under the process and maps to a unique mixed state $\eta(w) \in \Delta^2$ via the iterated MSP update (1).

The Blackwell measure μ_B . When a transformer is trained on sequences sampled from the process, each sequence w appears in proportion to its true probability $P(w)$. The resulting distribution over mixed states is

$$\mu_B(\eta) = \sum_{w: \eta(w)=\eta} P(w). \quad (24)$$

μ_B is the *stationary* measure of the MSP: the unique probability measure on $\mathcal{R}$ left invariant by the IFS dynamics. In practice, the empirical histogram over mixed states from N sampled sequences converges to μ_B , which weights states by how often they are visited, not by how many geometrically distinct states exist.

The uniform-unique measure μ_{uniq} . Enumerating the set $\mathcal{R}_n$ of all geometrically distinct mixed states reachable by length- n sequences and assigning equal weight to each,

$$\mu_{\text{uniq}}(\eta) = \frac{1}{|\mathcal{R}_n|}, \quad \eta \in \mathcal{R}_n, \quad (25)$$

gives a distribution that spreads mass across the entire attractor, including interior points μ_B almost ignores.

A.2 Missing Mass

Definition A.1 (Missing mass). Given N sampled sequences, the *missing mass* is the total process probability assigned to words whose mixed state was never visited:

$$M_N = \sum_{\substack{w \in \mathcal{W}_n \\ \hat{c}(w)=0}} P(w), \quad (26)$$

where $\hat{c}(w)$ is the empirical count of word w .

By the Good–Turing estimator, $\mathbb{E}[M_N] \approx N_1/N$ where N_1 is the number of words seen exactly once.

Remark A.2 (Heuristic scaling estimate for M_N). The IFS maps $f^{(z)}$ are contractions on Δ^2 (rigorously, in the Hilbert projective metric, by Birkhoff’s theorem, unconditionally on strict positivity of $T^{(z)}$; see Section 8). Treating the corner e_A as an *approximate* fixed point of $f^{(0)}$ for illustration (only exact in the degenerate limits $x \rightarrow 0$ or $\alpha \rightarrow 1$; see the caveat below), the Euclidean derivative there is $\partial f_A^{(0)} / \partial \eta_A|_{\eta_A=1} = p_{\text{off}}/p_{\text{dom}} =: \kappa$, and using $p_{\text{dom}} + 2p_{\text{off}} = 1$ together with $\Delta p = p_{\text{dom}} - p_{\text{off}}$,

$$\kappa = \frac{p_{\text{off}}}{p_{\text{dom}}} = \frac{1 - \Delta p}{1 + 2\Delta p}. \quad (27)$$

A word of length n reaching an interior mixed state must deviate from its dominant symbol at each step; treating each such deviation as costing a factor $\sim \kappa$ (an order-of-magnitude, not exact, statement) gives the asymptotic estimate

$$M_N \sim \mathcal{O}\left(\left(\frac{p_{\text{off}}}{p_{\text{dom}}}\right)^n\right) = \mathcal{O}\left(\left(\frac{1 - \Delta p}{1 + 2\Delta p}\right)^n\right). \quad (28)$$

Caveat (why this is only heuristic). e_A is an exact fixed point of $f^{(0)}$ only when $x \rightarrow 0$ or $\alpha \rightarrow 1$; in general $f^{(0)}(e_A)_A = \alpha y/p_{\text{dom}} \neq 1$. And - more importantly - the corner is even the right *direction* of attraction only

on regime (I) ($\alpha > \frac{1}{3}, x < \frac{1}{3}$): Appendix B shows by direct computation that on the other branch of $\Delta p > 0$, $f^{(0)}(\eta)_A$ is *decreasing* in η_A , so e_A behaves as a repeller there, and $\kappa = p_{\text{off}}/p_{\text{dom}}$ does not describe the dynamics at all. This estimate is therefore stated only for regime (I).

A.3 Standard Deviations and the Lyapunov Exponent

Let η_1 denote the first coordinate of a mixed state, and $\sigma_{\text{seq}}^2 = \text{Var}_{\mu_B}(\eta_1)$, $\sigma_{\text{uniq}}^2 = \text{Var}_{\mu_{\text{uniq}}}(\eta_1)$.

The IFS $\{f^{(z)}\}$ generates a random dynamical system in which, at each belief η , symbol z is chosen with probability $\Pr(z | \eta) = \eta T^{(z)} \mathbf{1}$ (the process's own emission law, not uniformly among $\{0, 1, 2\}$). The correctly-weighted (quenched) Lyapunov exponent [3] is

$$\lambda_{\text{Lyap}} = \int_{\Delta^2} \sum_{z \in \{0,1,2\}} \Pr(z | \eta) \log \left| \frac{\partial f^{(z)}(\eta)}{\partial \eta} \right| d\mu_B(\eta) < 0 \quad (29)$$

(negative since each $f^{(z)}$ contracts on regime (I)). The heuristic scaling argument $\sigma_{\text{seq}} \sim e^{n\lambda_{\text{Lyap}}}$ (shrinking support diameter after n steps) against $\sigma_{\text{uniq}} \sim \text{const}$ (the attractor spans a fixed region regardless of n) gives

$$\frac{\sigma_{\text{uniq}}^2}{\sigma_{\text{seq}}^2} \sim e^{2n|\lambda_{\text{Lyap}}|} \rightarrow \infty, \quad (30)$$

consistent with μ_B concentrating at corners while the set of distinct reachable states spreads over the interior.

B Corner Concentration in the Blackwell Measure

B.1 Scope: $\Delta p > 0$ is not enough

By Eq. (6), $\Delta p > 0$ holds on two disjoint branches: (I) $\alpha > \frac{1}{3}, x < \frac{1}{3}$, or (II) $\alpha < \frac{1}{3}, x > \frac{1}{3}$. These are qualitatively different. Take $z = 0$, dominant state A : a direct computation gives

$$f^{(0)}(\eta)_A = \frac{\alpha[x + (1-3x)\eta_A]}{p_{\text{off}} + \Delta p \eta_A}.$$

Differentiating, the numerator of $\partial f_A^{(0)} / \partial \eta_A$ simplifies (substituting $p_{\text{off}} = \alpha x + b(1-x)$, $\Delta p = (3\alpha - 1)(1-3x)/2$, and cancelling) to exactly

$$\alpha(1-3x)p_{\text{off}} - \alpha x \Delta p = \alpha(1-3x)b, \quad (31)$$

which is positive iff $x < \frac{1}{3}$ (since $b = (1-\alpha)/2 > 0$ for every $\alpha \in (0, 1)$) - *independent of whether $\alpha > \frac{1}{3}$ or $\alpha < \frac{1}{3}$* . So on branch (II), where $x > \frac{1}{3}$, Eq. (31) is *negative*: $f^{(0)}(\eta)_A$ is *decreasing* in η_A . Direct numerical check confirms this: at $(\alpha, x) = (0.2, 0.4)$ (so $\Delta p = 0.04 > 0$), $f^{(0)}(\eta)_A$ falls from 0.242 to 0.117 as η_A increases from 0.05 to 0.95. Everything below is therefore restricted to **regime (I)**; this is also why Section 7's parameter point, which has $\Delta p < 0$, is explicitly out of scope for this appendix.

B.2 Emission Probability as a Linear Function of Belief

For Mess3, the probability of emitting symbol z given belief η is

$$\Pr(z | \eta) = \eta T^{(z)} \mathbf{1} = p_{\text{off}} + \Delta p \cdot \eta_{s^*(z)}, \quad (32)$$

where $s^*(z)$ is the dominant state for z . Directly,

$$\Pr(z | \eta) = p_{\text{dom}} \eta_{s^*(z)} + p_{\text{off}}(1 - \eta_{s^*(z)}) = p_{\text{off}} + \Delta p \eta_{s^*(z)}.$$

On regime (I) ($\Delta p > 0$), this is strictly increasing in $\eta_{s^*(z)}$, equals p_{dom} at the corner $\eta = e_{s^*(z)}$ (its maximum), and equals $\frac{1}{3}$ at $\eta_{s^*(z)} = \frac{1}{3}$ - in particular at $\eta = \pi$, since $p_{\text{off}} + \Delta p/3 = (2p_{\text{off}} + p_{\text{dom}})/3 = 1/3$.

B.3 Monotonicity Lemma

Lemma B.1. *On regime (I) ($\alpha > \frac{1}{3}$, $x < \frac{1}{3}$), $f^{(z)}$ is strictly monotone increasing in $\eta_{s^*(z)}$, and for any $\eta \in \Delta^2$,*

$$f^{(z)}(\eta)_{s^*(z)} \geq \frac{\alpha x}{p_{\text{off}}}, \quad (33)$$

this lower bound exceeding $\frac{1}{3}$ exactly when $\alpha > \frac{1-x}{1+3x}$ - a strictly stronger condition than $\alpha > \frac{1}{3}$ for every $x \in (0, \frac{1}{3})$.

Proof. Monotonicity is Eq. (31), positive on regime (I). Since $f^{(0)}(\eta)_A$ is then increasing in η_A , its minimum over Δ^2 is at $\eta_A = 0$:

$$\min_{\eta \in \Delta^2} f^{(0)}(\eta)_A = f^{(0)}(e_B \text{ or } e_C)_A = \frac{\alpha x}{p_{\text{off}}}.$$

The condition $\alpha x / p_{\text{off}} > \frac{1}{3}$ is equivalent, in turn, to

$$\begin{aligned} 3\alpha x > p_{\text{off}} = \alpha x + b(1-x) &\iff 2\alpha x > b(1-x) = \frac{(1-\alpha)(1-x)}{2} \iff 4\alpha x > (1-\alpha)(1-x) \\ &\iff \alpha(1+3x) > 1-x \iff \alpha > \frac{1-x}{1+3x}. \end{aligned}$$

□

Remark B.2 (The $> \frac{1}{3}$ conclusion is not automatic on regime (I)). The threshold $\alpha > (1-x)/(1+3x)$ fails at several points inside regime (I), e.g. $(\alpha, x) = (0.6, 0.05)$ (need $\alpha > 0.826$) and $(\alpha, x) = (0.9, 0.02)$ (need $\alpha > 0.9245$; the given $\alpha = 0.9$ falls short, giving $\alpha x / p_{\text{off}} \approx 0.269 < \frac{1}{3}$). It holds, by contrast, at $(\alpha, x) = (0.6, 0.15)$ (≈ 0.346 , just above) and comfortably at $(\alpha, x) = (0.85, 0.28)$ (≈ 0.815). Monotonicity itself (the qualitative push toward the corner) is unaffected by this; only the specific numeric guarantee “more than a third of the mass” is.

B.4 Interior Mass Bound

Proposition B.3 (Interior has low μ_B -mass). *On regime (I), for $\mathcal{I}_\delta = \{\eta \in \Delta^2 : \min_i \eta_i \geq \delta\}$,*

$$\mu_B(\mathcal{I}_\delta) \lesssim \left(\frac{p_{\text{off}}}{p_{\text{dom}}} \right)^{k(\delta)} = \rho^{-k(\delta)}, \quad (34)$$

where $k(\delta)$ is (order of magnitude) the number of symbol switches needed to reach $\mathcal{I}_\delta$ from a corner.

We state this as a heuristic proposition, not a theorem, because its supporting argument treats a belief “near” a corner as if it were exactly at the corner ($\Pr(z' \mid \eta) \approx p_{\text{off}} < p_{\text{dom}} \approx \Pr(z \mid \eta)$) and does not quantify the approach rate exactly; both approximations become exact only as $\eta \rightarrow e_{s^*(z)}$. We did not find a counterexample to the conclusion, only to the rigor of the given argument for it, which mirrors the missing-mass estimate of Appendix A in character.

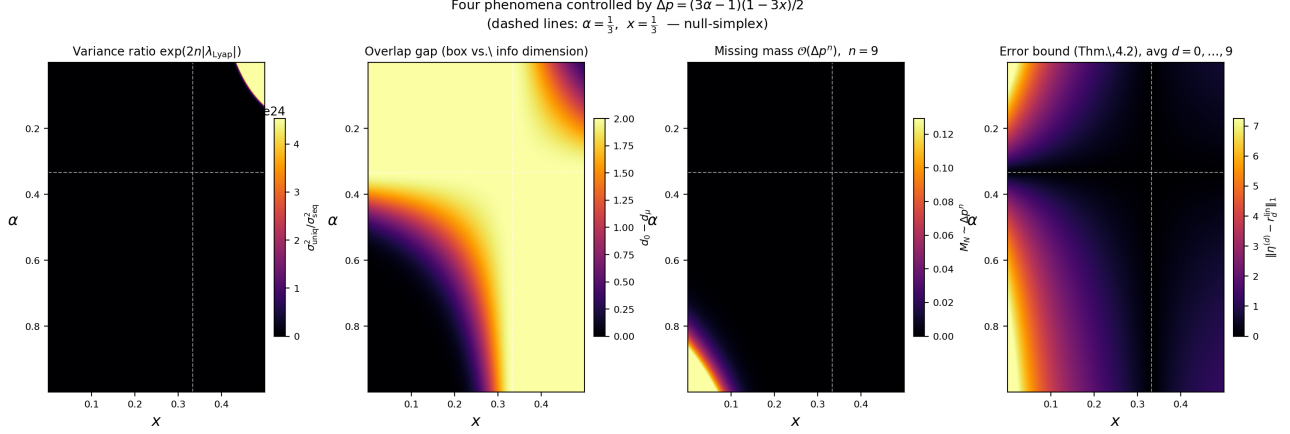


Figure 4: All four phenomena are visualised as heatmaps over the Mess3 parameter space (x, α) , $n = 9$. From left to right: (i) variance ratio $\sigma_{\text{uniq}}^2/\sigma_{\text{seq}}^2 \sim e^{2n|\lambda_{\text{Lyap}}|}$ (Appendix A); (ii) overlap gap $d_0 - d_\mu$ from [3]; (iii) missing mass $M_N \sim \mathcal{O}((p_{\text{off}}/p_{\text{dom}})^n)$ (Appendix A); (iv) the error bound of Eq. (15), averaged over $d = 0, \dots, 9$. Dashed lines mark $\alpha = \frac{1}{3}$ and $x = \frac{1}{3}$, where all four quantities vanish or are maximised.

C The Closed Form for T and Its Consequences

C.1 Derivation of Eq. (7)

Row A of $T = \sum_z T^{(z)}$ is the sum of row A of $T^{(0)}$, row A of $T^{(1)} = \text{cyc}(T^{(0)}, 1)$, and row A of $T^{(2)} = \text{cyc}(T^{(0)}, 2)$ (cyclic permutation of both rows and columns of Eq. (3)). Writing these out,

$$\text{row } A \text{ of } T^{(0)} = (\alpha y, bx, bx), \quad \text{row } A \text{ of } T^{(1)} = (by, \alpha x, bx), \quad \text{row } A \text{ of } T^{(2)} = (by, bx, \alpha x).$$

Summing coordinatewise, using $\alpha + 2b = \alpha + (1 - \alpha) = 1$:

$$\alpha y + by + by = y(\alpha + 2b) = y = 1 - 2x, \quad bx + \alpha x + bx = x(\alpha + 2b) = x.$$

So row A of T is $(1 - 2x, x, x)$, independent of α ; by the $\mathbb{Z}_3$ symmetry the same holds (cyclically) for rows B, C , giving exactly Eq. (7).

This closed form is used in Appendix D (the exact scalar action of T on zero-sum vectors, Eq. (8)) and in Section 6 (collapsing internal T 's in the higher-order residual expansion).

D The Multi-Step Error Bound: Proof Attempt

This appendix carries out the induction sketched in Section 5.2.

D.1 Telescoping

Define $\Phi_i := \eta^{(i)} T^{d-i}$, $\Psi_i := r_i^{\text{lin}} T^{d-i}$ for $0 \leq i \leq d$. Since $\eta^{(0)} = r_0^{\text{lin}} = \pi$ and $\pi T^k = \pi$, $\Phi_0 = \Psi_0 = \pi$; and $\Phi_d = \eta^{(d)}$, $\Psi_d = r_d^{\text{lin}}$, so

$$\eta^{(d)} - r_d^{\text{lin}} = \sum_{i=1}^d [(\Phi_i - \Psi_i) - (\Phi_{i-1} - \Psi_{i-1})] = \sum_{i=1}^d \varepsilon_i T^{d-i},$$

where, using $\eta^{(i)} = \eta^{(i-1)} T^{(z_i)} / P(z_i | \eta^{(i-1)})$ and $r_i^{\text{lin}} = r_{i-1}^{\text{lin}}(T + \Delta_i)$,

$$\varepsilon_i := \frac{\eta^{(i-1)} T^{(z_i)}}{P(z_i | \eta^{(i-1)})} - \eta^{(i-1)} T - r_{i-1}^{\text{lin}} \Delta_i.$$

Adding and subtracting $\hat{\eta}^{(i)} := \eta^{(i-1)} S^{(z_i)} = \eta^{(i-1)} T + \eta^{(i-1)} \Delta_i$,

$$\varepsilon_i = \underbrace{\left[\eta^{(i)} - \hat{\eta}^{(i)} \right]}_{=: \delta_i} + \underbrace{\left(\eta^{(i-1)} - r_{i-1}^{\text{lin}} \right) \Delta_i}_{\text{starting-point term}}. \quad (35)$$

D.2 The normalization error δ_i is bounded for every i , not only $i = 1$

Write $w_{jk} := \eta_j^{(i-1)} T_{jk}^{(z_i)} \geq 0$ and $v_k := \sum_j w_{jk}$. Then $\eta_k^{(i)} = v_k / P(z_i | \eta^{(i-1)})$, and since $P(z_i | \eta^{(i-1)}) = \sum_j \eta_j^{(i-1)} P(z_i | j)$ is a convex combination of values in $\{p_{\text{off}}, p_{\text{dom}}\}$, $\eta_k^{(i)} \in [v_k / p_{\text{dom}}, v_k / p_{\text{off}}]$. Separately, $\hat{\eta}_k^{(i)} = \sum_j w_{jk} / P(z_i | j)$ with each $P(z_i | j) \in \{p_{\text{off}}, p_{\text{dom}}\}$, so term-by-term $\hat{\eta}_k^{(i)} \in [v_k / p_{\text{dom}}, v_k / p_{\text{off}}]$ too. Both landing in the same band gives mutual domination $\eta_k^{(i)} / \hat{\eta}_k^{(i)} \in [1/\rho, \rho]$ for every k , and since both $\eta^{(i)}, \hat{\eta}^{(i)}$ are probability vectors, the TV argument from Lemma 5.1 applies verbatim:

$$\|\delta_i\|_1 \leq \frac{2(\rho - 1)}{\rho} \quad \text{for every } i = 1, \dots, d. \quad (36)$$

(This already generalizes Lemma 5.1, whose proof was the special case $i = 1$, $\eta^{(0)} = \pi$; the argument above did not use $\eta^{(i-1)} = \pi$ anywhere, only that $\eta^{(i-1)}$ is a probability vector.)

D.3 The starting-point term

By Cauchy–Schwarz-type submultiplicativity,

$$\|(\eta^{(i-1)} - r_{i-1}^{\text{lin}}) \Delta_i\|_1 \leq \|\eta^{(i-1)} - r_{i-1}^{\text{lin}}\|_1 \cdot \|\Delta_i\|_{1 \rightarrow 1}, \quad (37)$$

where $\|\Delta_i\|_{1 \rightarrow 1} = \max_s \sum_d |\Delta_{i, sd}|$. For the induction to close at the same constant as Eq. (15) - i.e. for $\|\varepsilon_i\|_1$ to be bounded by $2(\rho - 1)/\rho$ independent of i , matching Eq. (36) - one would want

$$\|\Delta_i\|_{1 \rightarrow 1} \leq \frac{\rho - 1}{\rho}. \quad (38)$$

This does not hold. Both $S^{(z_i)}$ and T are row-stochastic, so $\Delta_i = S^{(z_i)} - T$ has zero row sums and $\|\Delta_i\|_{1 \rightarrow 1} = \max_s \sum_d |S_{sd}^{(z_i)} - T_{sd}|$ can be computed directly from Eqs. (3) and (7). At $(\alpha, x) = (0.85, 0.28)$, where $p_{\text{dom}} \approx 0.416$, $p_{\text{off}} \approx 0.292$, $\rho \approx 1.425$, the right-hand side of Eq. (38) is

$$\frac{\rho - 1}{\rho} \approx 0.298, \quad (39)$$

while direct computation of Δ_i 's row-sums of absolute values gives a maximum of ≈ 1.070 - larger by a factor of ≈ 3.6 , and still larger than $2(\rho - 1)/\rho \approx 0.596$. Nor is this a function of ρ alone: at $(\alpha, x) = (0.6, 0.15)$, $\rho \approx 1.846$ is larger than at $(0.85, 0.28)$, yet $\|\Delta_i\|_{1 \rightarrow 1} \approx 0.392$ is smaller. So $\|\Delta_i\|_{1 \rightarrow 1}$ depends on x independently of $\Delta\rho$, and no bound of the form $c \cdot (\rho - 1)/\rho$ holds uniformly over the parameter space.

D.4 Where this leaves the induction

Falling back on the trivial bounds $\|\Delta_i\|_{1 \rightarrow 1} \leq 2$ and $\|\eta^{(i-1)} - r_{i-1}^{\text{lin}}\|_1 \leq 2$ (both hold since these are differences of probability vectors), Eq. (35) gives $\|\varepsilon_i\|_1 \leq \frac{2(\rho-1)}{\rho} + 4$, and hence a valid bound

$$\|\eta^{(d)} - r_d^{\text{lin}}\|_1 \leq \left(\frac{2(\rho - 1)}{\rho} + 4 \right) \cdot \frac{1 - (1 - 3x)^d}{3x}, \quad (40)$$

which is too weak to be useful (its right-hand side exceeds 4, while the left-hand side, a distance between probability vectors, is at most 2). Eq. (15) with the tight constant $2(\rho - 1)/\rho$ is therefore a heuristic. The rest of the telescoping argument works out - T acts as the precise scalar $1 - 3x$ on zero-sum vectors (Eq. (8)), so $\|\varepsilon_i T^{d-i}\|_1 = (1 - 3x)^{d-i} \|\varepsilon_i\|_1$; the gap is entirely in bounding $\|\varepsilon_i\|_1$ itself for $i > 1$.

E Experiment Details

All experiments use a 1-layer transformer with $d_{\text{model}} = 64$, context length 100, trained on 2×10^6 sequences from Mess3 with cross-entropy loss. Default parameters: $x = 0.15$, $\alpha = 0.6$ (regime (I): $\alpha > \frac{1}{3}$, $x < \frac{1}{3}$). SGD: learning rate 10^{-2} , no momentum. Adam: default $\beta_1 = 0.9$, $\beta_2 = 0.999$.

F Training Plots

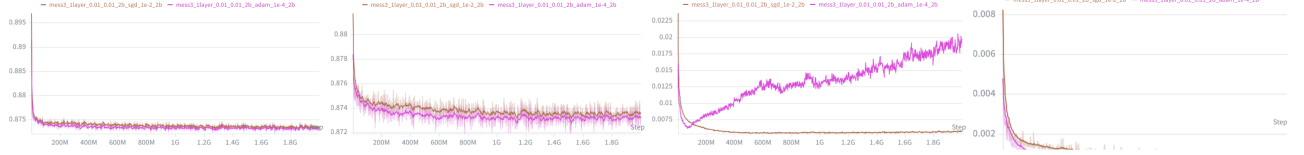


Figure 5: Training metrics for the model at $(x, \alpha) = (0.01, 0.01)$. From left to right: Train Loss, Test Loss, Simplex MSE, and KL Divergence over training steps.

G LLM Usage Declaration

Claude Sonnet 5 and 4.6 (High) contributed to the heuristic claims (error bounds), to writing the Python scripts used to generate Figure 4, and to refining L^AT_EX formatting. Gemini 3 Pro was also used for formatting.